

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, C++, Java, SQL, Bash

LLM & Agents

LLM agents, tool use, output verification, evaluation, prompt design, OpenAI / Anthropic / Groq APIs, Ollama, MCP

Retrieval & RAG

RAG, RRF, cross-encoder reranking, MMR, sentence-transformers, FAISS, cell-level provenance

ML & Healthcare

PyTorch, JAX, HuggingFace Transformers, NumPy, transfer learning, medical-imaging classification, clinical NLP

Backend & Cloud

FastAPI, Node.js, REST APIs, async services, Docker, AWS, GCP, Azure, Git, GitHub Actions

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label and curate data for production clinical models; cut manual work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered a multi-tenant **NHS clinical eTriage** platform (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly patient submissions**, integrated with Mayden's iaptus clinical record platform; owned reliability and remediated **20+ VAPT findings** protecting sensitive patient data.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed **20+ REST APIs** (Node.js, Fastify, PostgreSQL) and integrated **LLM-powered** context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability through targeted refactoring.

Projects

AgenticSearch - Provenance-First LLM Agent

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **LLM agent** that plans typed retrieval, extracts entities with deterministic parsers before LLM fallback, **verifies outputs**, and returns **cell-level provenance** for every result, the kind of verifiable, auditable extraction that high-stakes domains require.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, and pipeline instrumentation; deployed as a live demo on **Google Cloud Run**.

KG2RAG-Enhanced - Multi-Hop Retrieval-Augmented QA

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama + LLaMA-3, sentence-transformers, llama-index, networkx

- Built the **multi-view retrieval** component of a team KG-guided **RAG** pipeline: query-side decomposition into sub-questions, per-view **dense retrieval**, and fusion via RRF, cross-encoder reranking, and MMR, lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG²RAG baseline on a **4,905-question** evaluation.
- Benchmarked the pipeline across 4,905-question and controlled N=1,000 runs with token and latency logging and 95% binomial confidence intervals.

AutoEval - Retrieval Evaluation Improvement Agent

github.com/Deva-1903/AutoEval

Python, IR evaluation tooling, Git automation

- Built an **evaluation-improvement agent** that proposes queries and labels, scores them against multiple systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added **anti-Goodharting guardrails** (schema checks, duplicate rejection, read-only holdout protection) and Git-backed auditable updates with report cards for a reproducible evaluation workflow.

refusal-decay - LLM Safety & Mechanistic Interpretability

github.com/Deva-1903/refusal-decay

Python, PyTorch, HuggingFace Transformers | COMPSCI 602 Research Methods, graded full marks

- Ran a **mechanistic-interpretability** study of how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct, extracting the residual-stream refusal direction via difference-in-means and designing causal interventions with random/orthogonal controls and a benign positive control; reported a clean null with bootstrap **95% confidence intervals** and McNemar's test rather than overclaiming.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification (healthcare AI research).
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.